

Impact of IRS in Multiple Operator mmWave Systems: Centralized vs. Distributed IRSs

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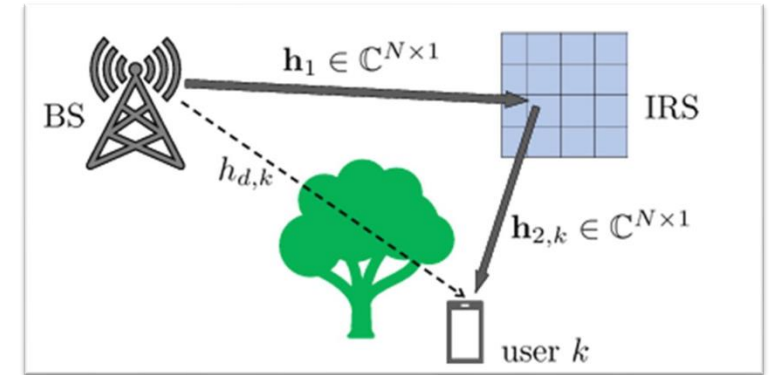


Introduction to IRS

- **Intelligent Reflecting Surfaces (IRS):** Meta surfaces with passive elements

- **Alters** the wireless channel as per our needs
- **Reflect** the signals in **specific directions**

- Perks: **Boosts SINR**, **energy efficiency**, **coverage**, **etc.**



- A hot research topic today: For e.g., IEEE has about **10K papers** in the **last 5 years**

Showing 1-25 of 10,463 results for ("All Metadata":Intelligent reflecting surfaces) OR ("All Metadata":Reconfigurable intelligent surfaces) ×

☐ Conferences (4,797)

☐ Journals (4,692)

☐ Early Access Articles (483)

☐ Magazines (450)

☐ Books (41)

- Many **unsolved problems still** exist! We will explore one very important problem today
- IRS in **Industry & standards**: ETSI, TSDSI wrkps, Qualcomm, ZTE (prototypes), etc.
 - <https://www.etsi.org/technologies/reconfigurable-intelligent-surfaces>

The Benchmark Performance using an IRS

Theorem 1 (Performance of the optimized IRS - beamforming configuration)

The rate obtained by the user k when the IRS is optimized to user k is

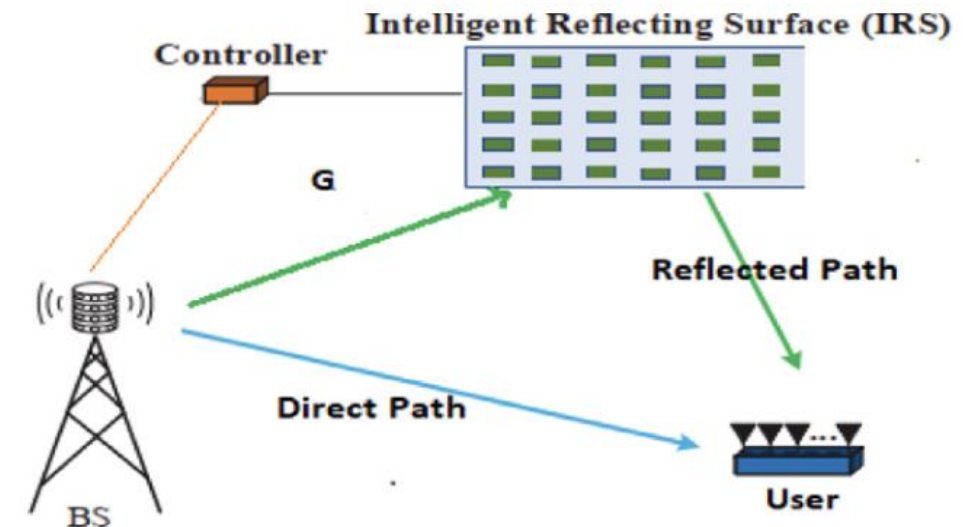
$$R_k^{BF} = \log_2 \left(1 + \frac{P}{\sigma^2} \left| \sqrt{\beta_{r,k}} \sum_{n=1}^N |h_{1,n}| |h_{2,k,n}| \times \exp(j \angle h_{d,k}) + \sqrt{\beta_{d,k}} h_{d,k} \right|^2 \right).$$

It is achieved when (due to *Cauchy - Schwarz inequality*),

$$\theta_{n,k}^{BF} = \angle h_{d,k} - \angle (h_{1,n} + h_{2,k,n}), \quad n = 1, \dots, N.$$

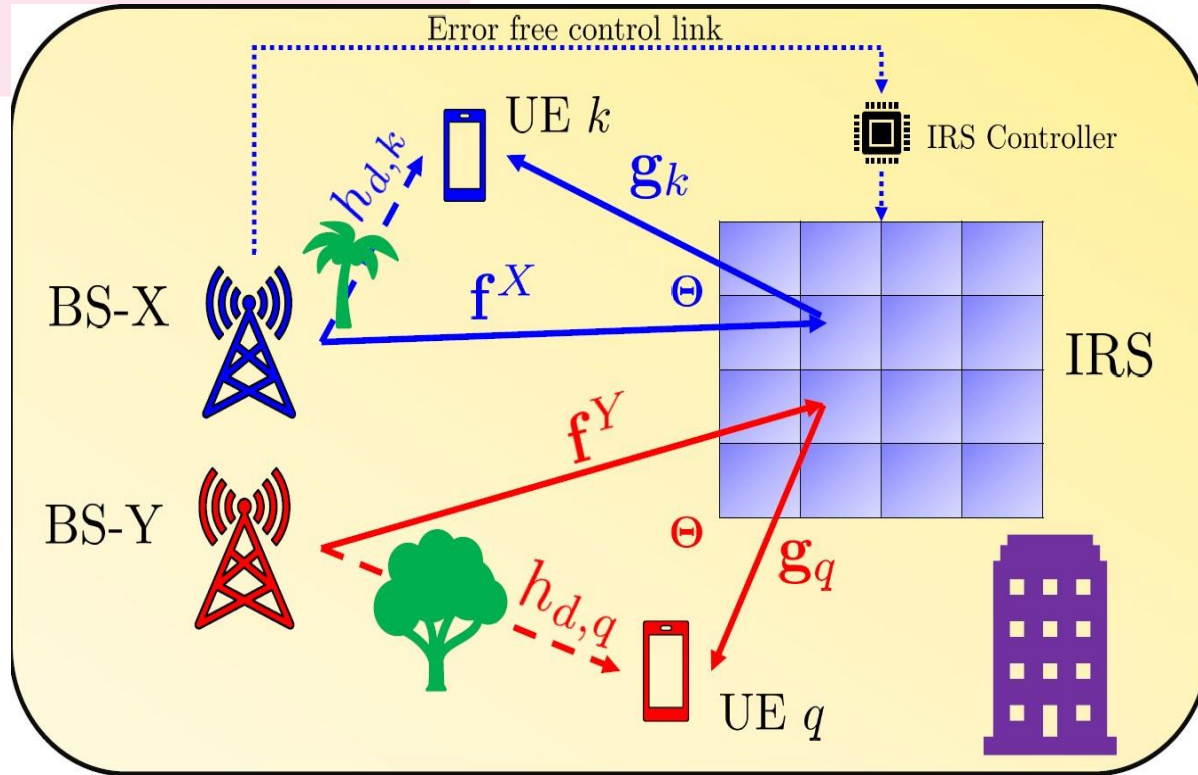
❑ State-of-the-art: "Three-fold overhead"

- CSI estimation - complexity of $O(N)$
- Expensive phase optimization @ BS
- Phase transport: BS to IRS - complexity of $O(N)$



Source: Google Images

Motivation for the problem

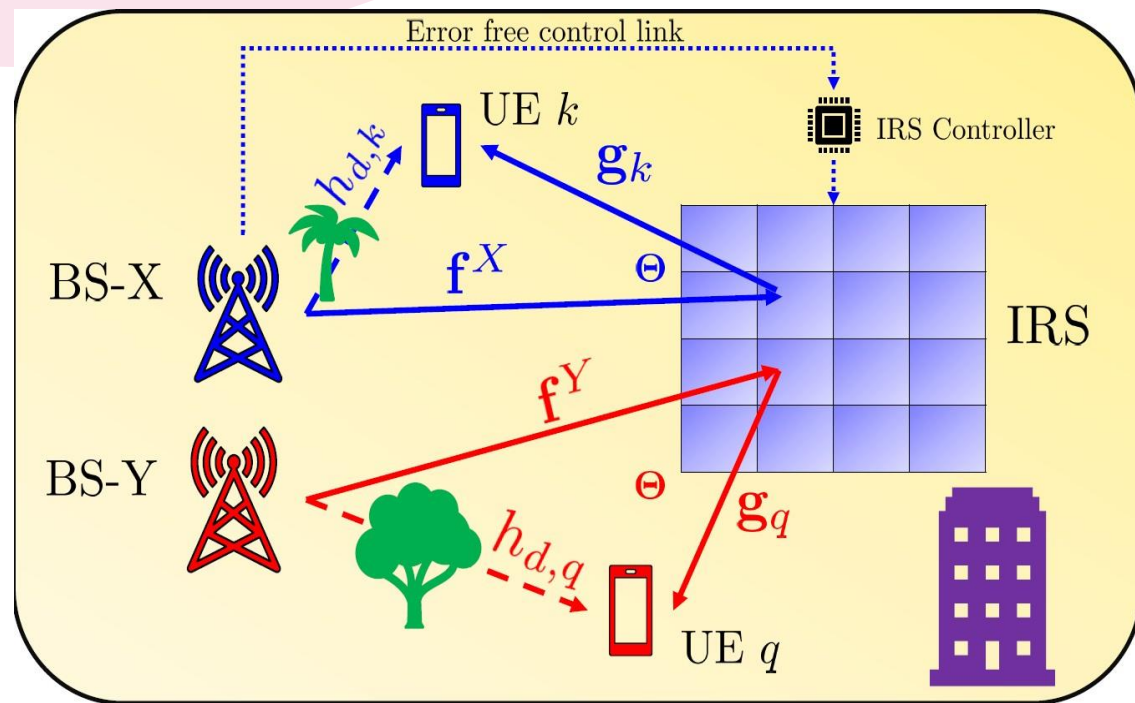


- 2 operators, X & Y (e.g., Airtel & Jio)
- IRS does not have a bandpass filter
- NO practical IRS can simultaneously beamform to UEs of BS - X & Y
 - Different channels @ in-band & OOB UEs
- Operator X controls the IRS
 - Optimal IRS @ in-band UEs
 - Random IRS @ OOB UEs

Does IRS Degrade Out-of-Band Performance in mmWave?

*For Sub-6 GHz: L. Yashvanth, Chandra. R. Murthy, "Does an IRS Degrade OOB Performance?" in SPAWC '23.

System Setting



- Downlink signal at k -th UE (served by BS-X):

$$y_k = (h_{d,k} + \mathbf{g}_k^T \Theta \mathbf{f}^X) x_k + n_k,$$

- Downlink signal at q -th UE (served by BS-Y):

$$y_q = (h_{d,q} + \mathbf{g}_q^T \Theta \mathbf{f}^Y) x_q + n_q,$$

➤ BS - X serves K UEs

➤ Operator X controls the IRS

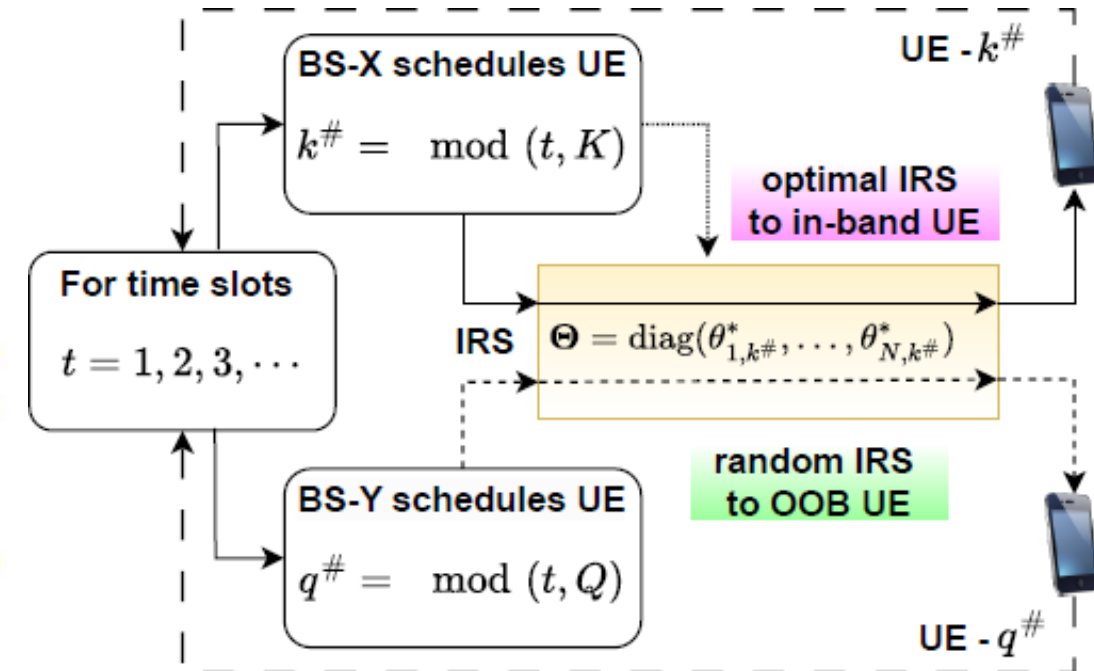
➤ BS - Y serves Q UEs

➤ Operator Y does not control the IRS

Scheduling Protocol: Round-Robin

Algorithm Protocol at BS-X and BS-Y

- 1: **Input:** UE indices at BS-X: $0, 1, \dots, K - 1$.
- 2: **Input:** UE indices at BS-Y: $0, 1, \dots, Q - 1$.
- 3: Initialize the UE index at BS-X to 0, at BS-Y to 0.
- 4: **repeat**
- 5: **for** time slot t **do**
- 6: $k^\# = \text{mod}(t, K)$. ▷ RR Scheduling at BS-X.
- 7: $q^\# = \text{mod}(t, Q)$. ▷ RR Scheduling at BS-Y.
- 8: Set $\Theta = \text{diag}(\theta_{1,k^\#}^*, \dots, \theta_{N,k^\#}^*)$ optimally.
- 9: ▷ Hence, a random Θ gets realized for OOB UEs.
- 10: Data transmission from BS-X to $k^\#$.
- 11: Data transmission from BS-Y to $q^\#$.
- 12: **end for**
- 13: **until** service time



OOB Performance in mm-Wave Bands

- ❑ FR-2 Bands in the 5G standards (24.25 GHz - 52.6 GHz, Rel. 15, 2018)
- ❑ **Sparse** channels with **few** propagation **paths**
- ❑ Channels are highly **directional**!



System Model in mmWave Frequency Bands

- Directional Saleh - Valenzuela model:

$$\mathbf{f}^p = \sqrt{\frac{N}{L_{1,p}}} \sum_{i=1}^{L_{1,p}} \gamma_{i,p}^{(1)} \mathbf{a}_N^*(\phi_{i,p}); \quad \mathbf{g}_\ell = \sqrt{\frac{N}{L_{2,\ell}}} \sum_{j=1}^{L_{2,\ell}} \gamma_{j,\ell}^{(2)} z \mathbf{a}_N^*(\psi_{j,\ell}), \quad p \in \{X, Y\}$$

- Array steering vector: $\mathbf{a}_N(\phi) = \frac{1}{\sqrt{N}} [1, e^{-j\pi\phi}, \dots, e^{-j(N-1)\pi\phi}]^T$

- Beam-resolution capability:

- N-element ULA can form at most N resolvable beams

- Resolvable beam book and angle book:

$$\mathcal{A} \triangleq \{\mathbf{a}_N(\phi), \phi \in \Phi\}; \quad \Phi \triangleq \left\{ \left(-1 + \frac{2i}{N} \right) \middle| i = 0, \dots, N-1 \right\}$$

- Uniform distribution over beam book:

$$\mathcal{U}_{\mathcal{A}}(\phi) = \frac{1}{|\Phi|} \mathbf{1}_{\{\phi \in \Phi\}} = \frac{1}{N} \mathbf{1}_{\{\phi \in \Phi\}}$$

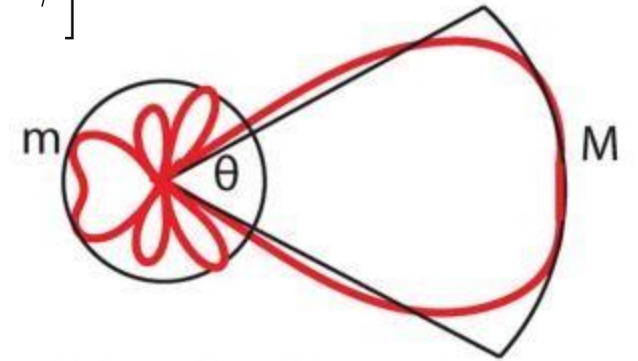
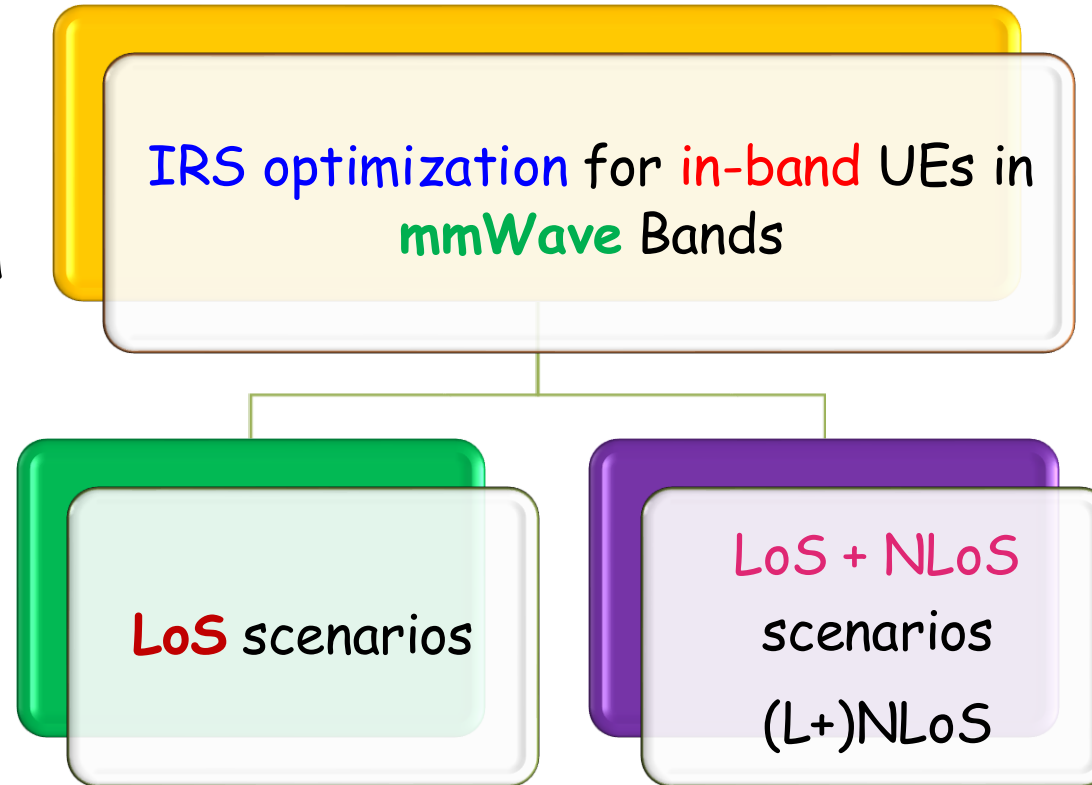


Fig. The "Flat-top" beam pattern @ IRS

Categories of the study in Centralized single IRS

LoS scenario

- ❖ IRS is **aligned** to the **dominant** cascaded path (virtual LoS)
- ❖ In-band UEs' **channel approximated** by the dominant path
- ❖ IRS vector is a **phasor**
- ❖ **Signaling** overhead does **NOT SCALE** with **N**



(L+)NLoS scenario

- ❖ IRS is **jointly aligned** to all cascaded paths
- ❖ **No structure** in the IRS vector
- ❖ **Signaling** overhead **scales linearly** with IRS elements

Channels for **OOB UEs** can have **more than one** spatial path

LoS Scenarios

- Channel simplification for dominant path:
$$h_k = N\gamma_{1,X}^{(1)}\gamma_{1,k}^{(2)}\mathbf{a}_N^H(\psi_{1,k})\mathbf{\Theta}\mathbf{a}_N^*(\phi_{1,X}) + h_{d,k}$$
$$\stackrel{(a)}{=} N\left(\gamma_{1,X}^{(1)}\gamma_{1,k}^{(2)}\left(\mathbf{a}_N^H(\phi_{1,X}) \odot \mathbf{a}_N^H(\psi_{1,k})\right)\right)\boldsymbol{\theta} + h_{d,k},$$
$$= N\gamma_{X,k}\dot{\mathbf{a}}_N^H(\omega_{X,k}^1)\boldsymbol{\theta} + h_{d,k}$$
- Optimal IRS vector
$$\boldsymbol{\theta}^{\text{opt}} = \frac{h_{d,k}\gamma_{X,k}^*}{|h_{d,k}\gamma_{X,k}|} \times N\dot{\mathbf{a}}_N(\omega_{X,k}^1)$$
- IRS has a **uni-directional** response
@ in-band channel angle
- With probability $\frac{L}{N}$, OOB UE **benefits!**
- With probability $1 - \frac{L}{N}$, OOB perf. **same** as without an IRS

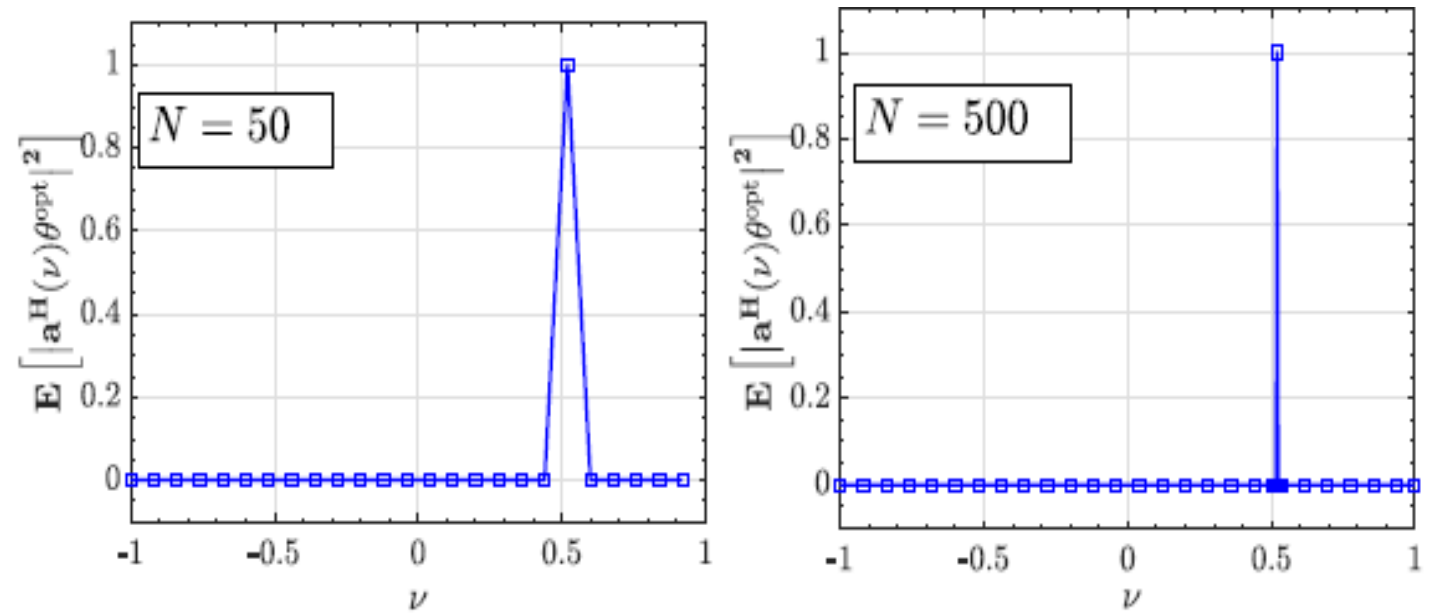


Fig. 3: Correlation response of the IRS vector and array steering vectors pointing at different spatial angles, ν , for (a) $N = 50$ and (b) $N = 500$. When $\nu = \omega_{X,k}^1$, the response attains its maximum value of 1.

Main Result - 1 (On Ergodic SE performance)

The ergodic sum SEs of operator $\mathbf{X}$ & $\mathbf{Y}$ scale as

$$\bar{S}_1^{(X)} \approx \frac{1}{K} \sum_{k=1}^K \log_2 \left(1 + \left[N^2 \beta_{r,k} + N \left(\frac{\pi^{3/2}}{4} \sqrt{\beta_{d,k} \beta_{r,k}} \right) + \beta_{d,k} \right] \frac{P}{\sigma^2} \right),$$

and

$$\bar{S}_1^{(Y)} \approx \frac{1}{Q} \sum_{q=1}^Q \left(\frac{\bar{L}}{N} \log_2 \left(1 + \left[\frac{N^2}{\bar{L}} \beta_{r,q} + \beta_{d,q} \right] \frac{P}{\sigma^2} \right) + \left(1 - \frac{\bar{L}}{N} \right) \log_2 \left(1 + \beta_{d,q} \frac{P}{\sigma^2} \right) \right),$$

where $\bar{L} \triangleq \min \{L, N\}$, and $L \triangleq L_1 L_2$

- ✓ IRS **does not degrade** the OOB performance! It occasionally helps the OOB UEs
- ✓ **Linear** scaling of SNR with N is guaranteed if $L \geq N$
- ✓ **Sub-linear** scaling of SNR with N if $L < N$.

The achievable OOB-SE in the **presence** of the IRS is **at least** the SE in the **absence** of the IRS

Numerical results for LoS scenarios

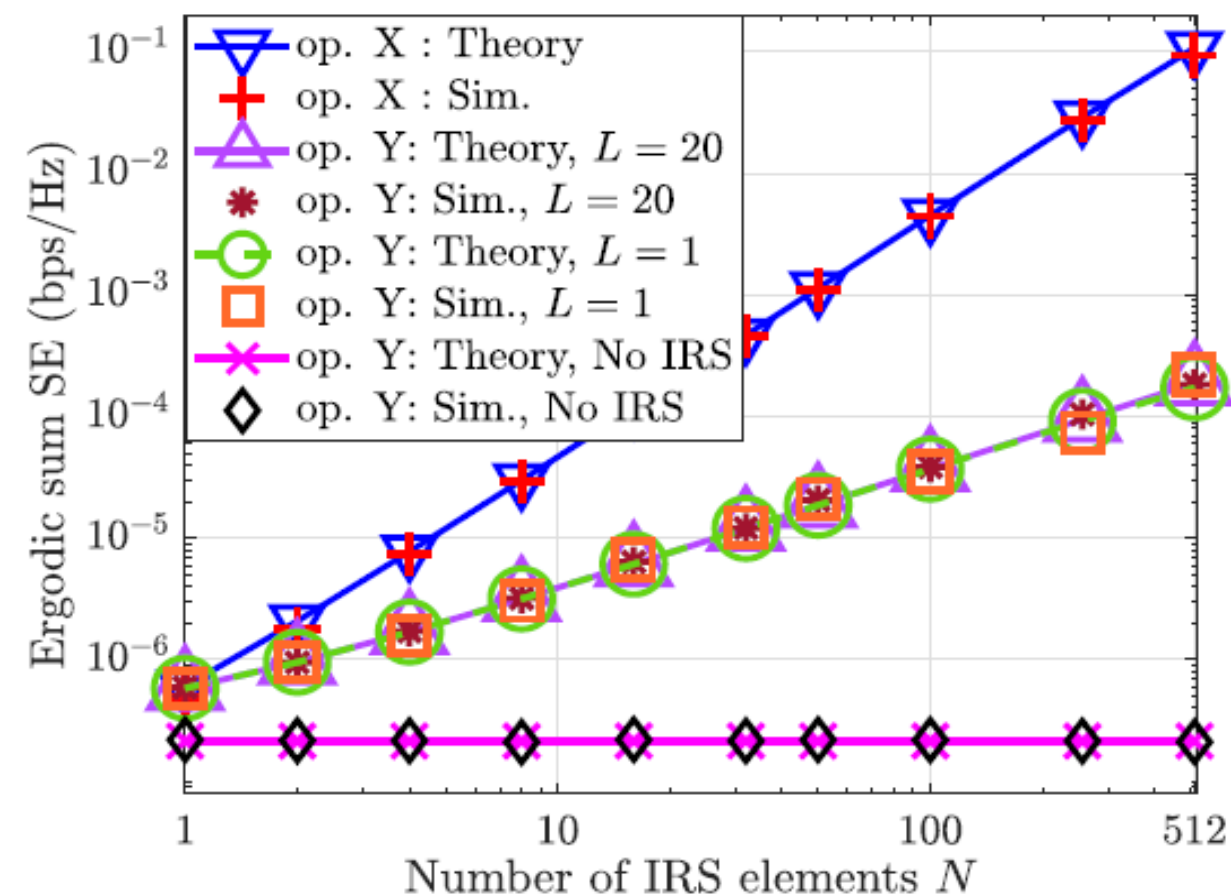


Fig. 6: Ergodic sum-SE vs. N in LoS scenarios at $\gamma = 70$ dB.

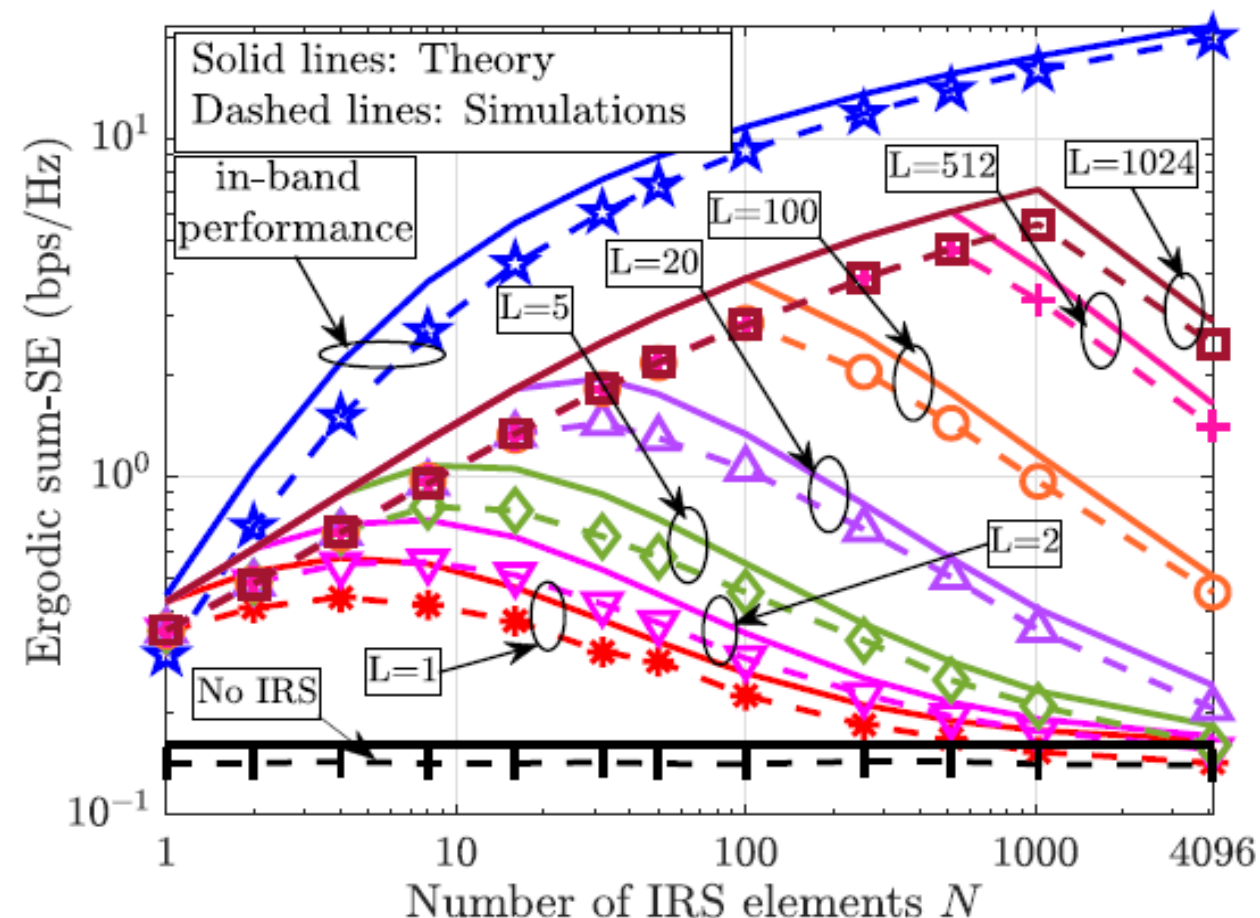


Fig. 7: Ergodic sum-SE vs. N in LoS scenarios at $\gamma = 130$ dB.

Main Results - 2 (On Instantaneous performance)

❑ Outage Probability:

$$F_{|h_q|^2}(\rho) \triangleq \Pr(|h_q|^2 < \rho) = 1 - e^{-\rho/\beta_{d,q}} - \frac{\bar{L}}{N} \left(\frac{\bar{L} e^{\frac{\bar{L}\beta_{d,q}}{N^2\beta_{r,q}}}}{N^2\beta_{r,q}} \mathcal{I}_0 \left(\rho; \beta_{d,q}, \frac{N^2}{\bar{L}}\beta_{r,q} \right) - e^{-\rho/\beta_{d,q}} \right),$$

where $\bar{L} = \min\{L, N\}$, and $\mathcal{I}_0(x; c_1, c_2) = \int_{c_1}^{\infty} e^{\left\{ -\left(\frac{x}{t} + \frac{t}{c_2} \right) \right\}} dt$.

❑ Channel quality gain indicator:

Define $G_1 = |h_q|^2$, and $G_0 = |h_q|^2$ in the *presence* and *absence* of the IRS, respectively. Then

$$\Pr(G_1 > \rho) \geq \Pr(G_0 > \rho) \quad \forall \rho \in \mathcal{R}^+,$$

The channel gain of the OOB-UE in the **presence** of an IRS **stochastically dominates** the channel gain in its **absence**.

Numerical results - 2 for LoS scenarios

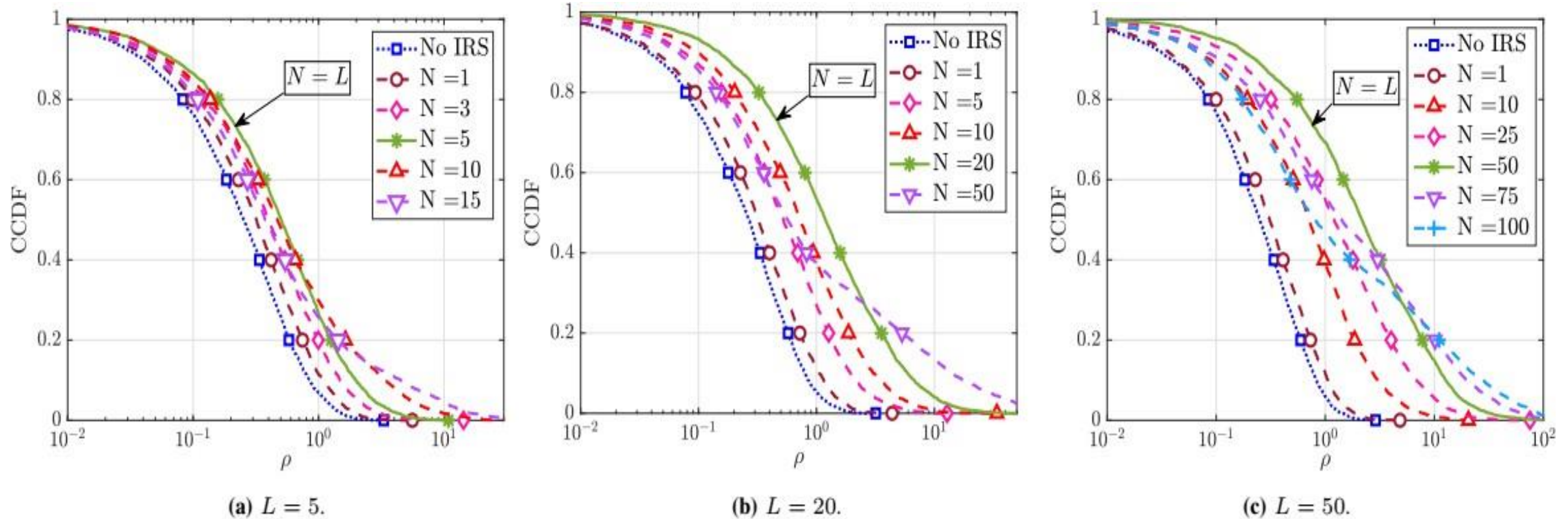


Fig. 8: CCDF of OOB UE's channel gain $|h_q|^2$ as a function of N in LoS scenarios of the mmWave bands at $\gamma = 130$ dB.

(L+)NLoS scenarios

- IRS is optimized to **LoS** and **NLoS** paths of the in-band UE's channel

- Channel: $h_k = h_{d,k} + \frac{N}{\sqrt{L}} \sum_{l=1}^L \gamma_{l,X}^{(1)} \gamma_{l,k}^{(2)} \mathbf{a}_N^H(\omega_{X,k}^l) \boldsymbol{\theta}$.

- Optimal IRS configuration:

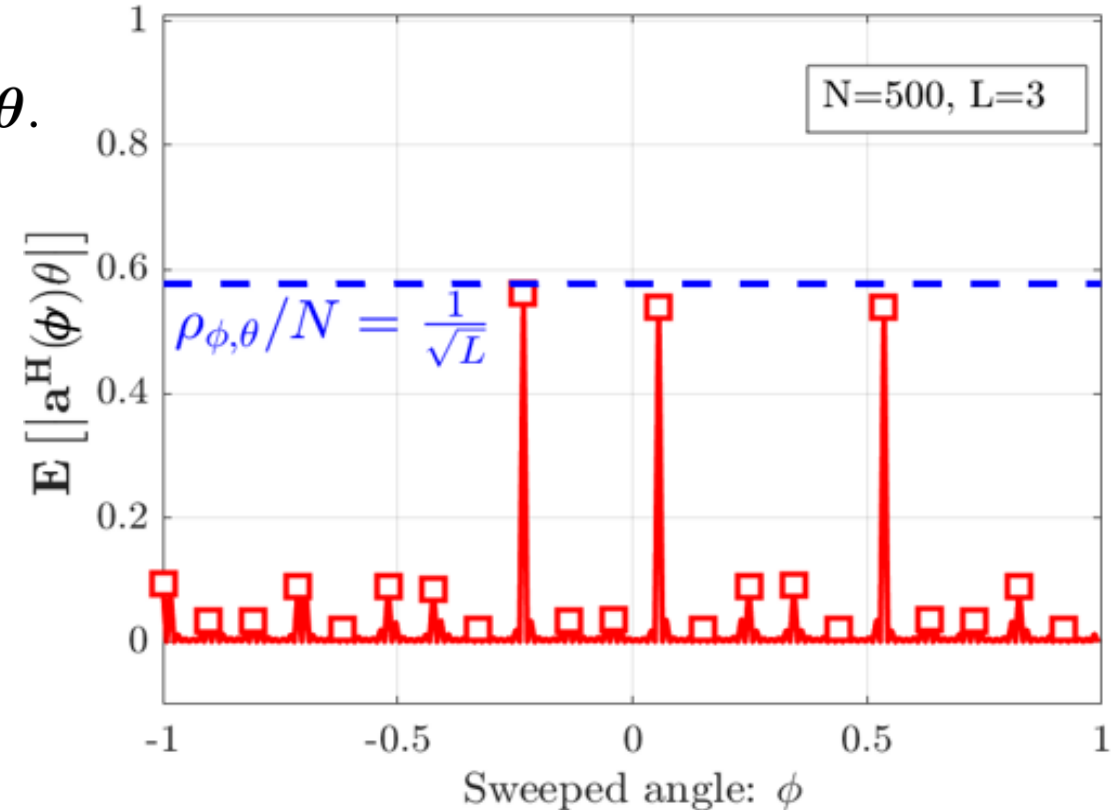
$$\boldsymbol{\theta}^{\text{opt}} = \frac{h_{d,k}}{|h_{d,k}|} \left(\sum_{l=1}^L \gamma_{l,X}^{(1)*} \gamma_{l,k}^{(2)*} \mathbf{a}_N(\omega_{X,k}^l) \right) \odot \frac{1}{\left| \sum_{l=1}^L \gamma_{l,X}^{(1)} \gamma_{l,k}^{(2)} \mathbf{a}_N(\omega_{X,k}^l) \right|},$$

- Directivity response:

Lemma 3. *The optimal IRS configuration*

has the following spatial amplitude response:

$$\rho_{\phi,\theta} = \begin{cases} \Omega \left(\frac{N}{\sqrt{L}} \right) + o(N), & \text{if } \phi \in \left\{ \omega_{X,k}^1, \dots, \omega_{X,k}^L \right\}, \\ o(N), & \text{if } \phi \in \Phi \setminus \left\{ \omega_{X,k}^1, \dots, \omega_{X,k}^L \right\}. \end{cases}$$



IRS distributes energy
without leakage

Main Result - 3 (On Ergodic performance)

The ergodic sum SEs of operator **X** & **Y** scale as

$$\bar{S}_L^{(X)} \approx \frac{1}{K} \sum_{k=1}^K \log_2 \left(1 + \left(N^2 \beta_{r,k} + N \sqrt{\pi \beta_{d,k} \beta_{r,k}} + \beta_{d,k} \right) \frac{P}{\sigma^2} \right),$$

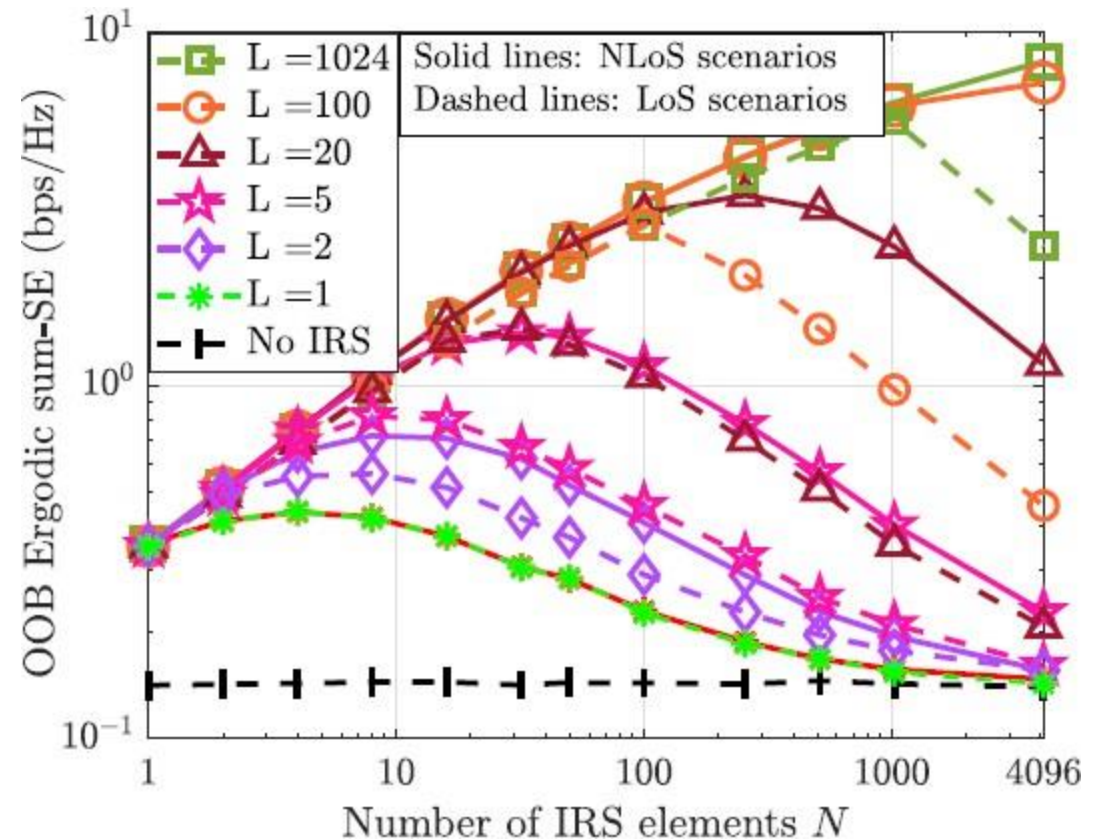
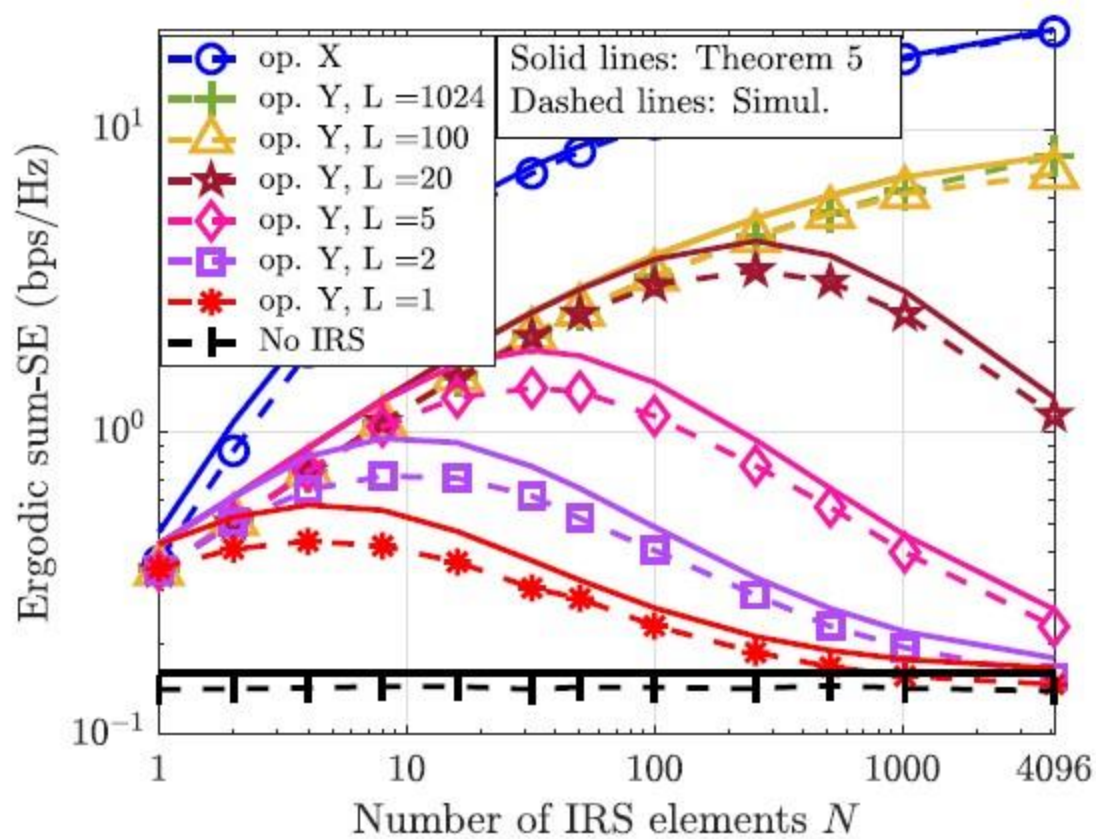
and

$$\bar{S}_L^{(Y)} \approx \begin{cases} \frac{1}{Q} \sum_{q=1}^Q \sum_{i=i_0}^L \frac{\binom{L}{i} \binom{N-L}{L-i}}{\binom{N}{L}} \times \log_2 \left(1 + \left(\beta_{d,q} + i \frac{N^2}{L^2} \beta_{r,q} \right) \frac{P}{\sigma^2} \right), & \text{if } L < N, \\ \frac{1}{Q} \sum_{q=1}^Q \log_2 \left(1 + (\beta_{d,q} + N \beta_{r,q}) \frac{P}{\sigma^2} \right), & \text{if } L \geq N. \end{cases}$$

where $i_0 \triangleq \max\{0, 2L - N\}$.

- ✓ IRS **does not degrade** the OOB performance
- ✓ **Linear** scaling of SNR with **N** is guaranteed if $L \geq \sqrt{N}$.
- ✓ **Sub-linear** scaling of SNR with **N** if $L < \sqrt{N}$.

Numerical results - 1 for (L+)NLoS scenarios



- Probability of 1-path alignment $\approx \frac{L^2}{N} > \frac{L}{N}$
- OOB perf. in NLOS is better than LOS
- Trade-off: More feedback @ in-band operator

Key Inferences:

Sufficient L for linear scaling in LoS: N

Sufficient L for linear scaling in (L+)NLoS: $\sqrt{N}$

OOB Perf. with Distributed IRSs

- A single IRS **benefits** the OOB UE with prob. $\frac{L}{N}$, & **does not affect** with prob. $1 - \frac{L}{N}$

- ❖ L : # paths via the IRS; N : # IRS elements

- Can we further boost OOB perf.? - Increase # paths

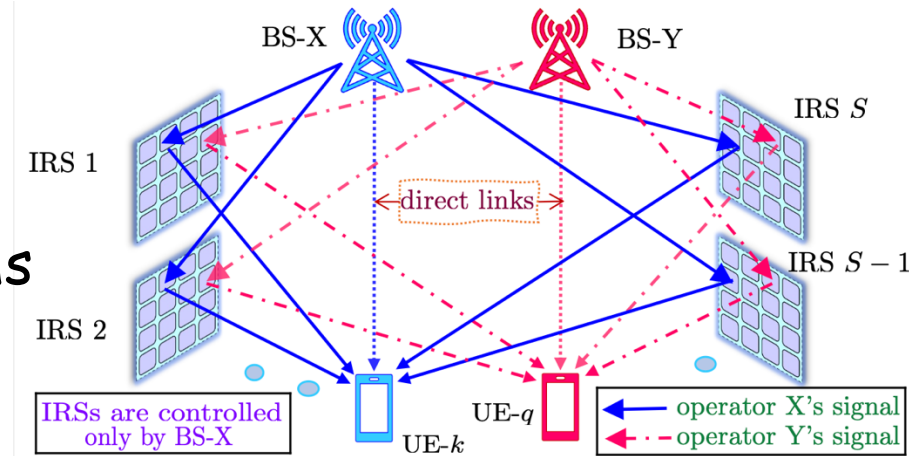
- ❖ **Solution** : Deploy multiple distributed IRSs

- ❖ Partition a N element IRS into S sub-IRSs with M elements : $SM = N$

- Dual benefits @ OOB UEs:

- ❖ Total # paths at OOB UE increases: $L \rightarrow SL$

- ❖ At an IRS, the probability of beam alignment increases: $\frac{L}{N} \rightarrow \frac{L}{M} = S \frac{L}{N}$



Main Result - 4: Ergodic Spectral Efficiency

➤ Ergodic sum-SE of the in-band operator - **X**:

$$\bar{S}_M^{(X)} = \frac{1}{K} \sum_{k=1}^K \langle S_k^{(X)} \rangle \approx \frac{1}{K} \sum_{k=1}^K \log_2 \left(1 + \left[N^2 \left(\frac{\pi^2}{16} + \eta \left(1 - \frac{\pi^2}{16} \right) \right) \beta_{r,k} + N \frac{\pi^{3/2}}{4} \sqrt{\beta_{d,k} \beta_{r,k} + \beta_{d,k}} \right] \frac{P}{\sigma^2} \right)$$

- The **in-band UE** achieves the **full $O(N^2)$ SNR** gain for any design of distributed IRSs

➤ Ergodic sum-SE of the out-of-band operator - **Y**:

$$\bar{S}_M^{(Y)} \approx \begin{cases} \frac{1}{Q} \sum_{q=1}^Q \sum_{s=0}^S \frac{S!}{(S-s)!s!} \left(\frac{L}{M} \right)^s \left(1 - \frac{L}{M} \right)^{S-s} \log_2 \left(1 + \left[\frac{sM^2}{L} \beta_{r,q} + \beta_{d,q} \right] \frac{P}{\sigma^2} \right), & \text{if } L < M, \\ \frac{1}{Q} \sum_{q=1}^Q \log_2 \left(1 + (\beta_{d,q} + N \beta_{r,q}) \frac{P}{\sigma^2} \right), & \text{if } L \geq M. \end{cases}$$

- The **OOB UE benefits** are governed by a **binomial distribution**
- How to achieve **$O(\log(N))$ OOB SE growth** with **probability 1?**

Sufficient # IRS?

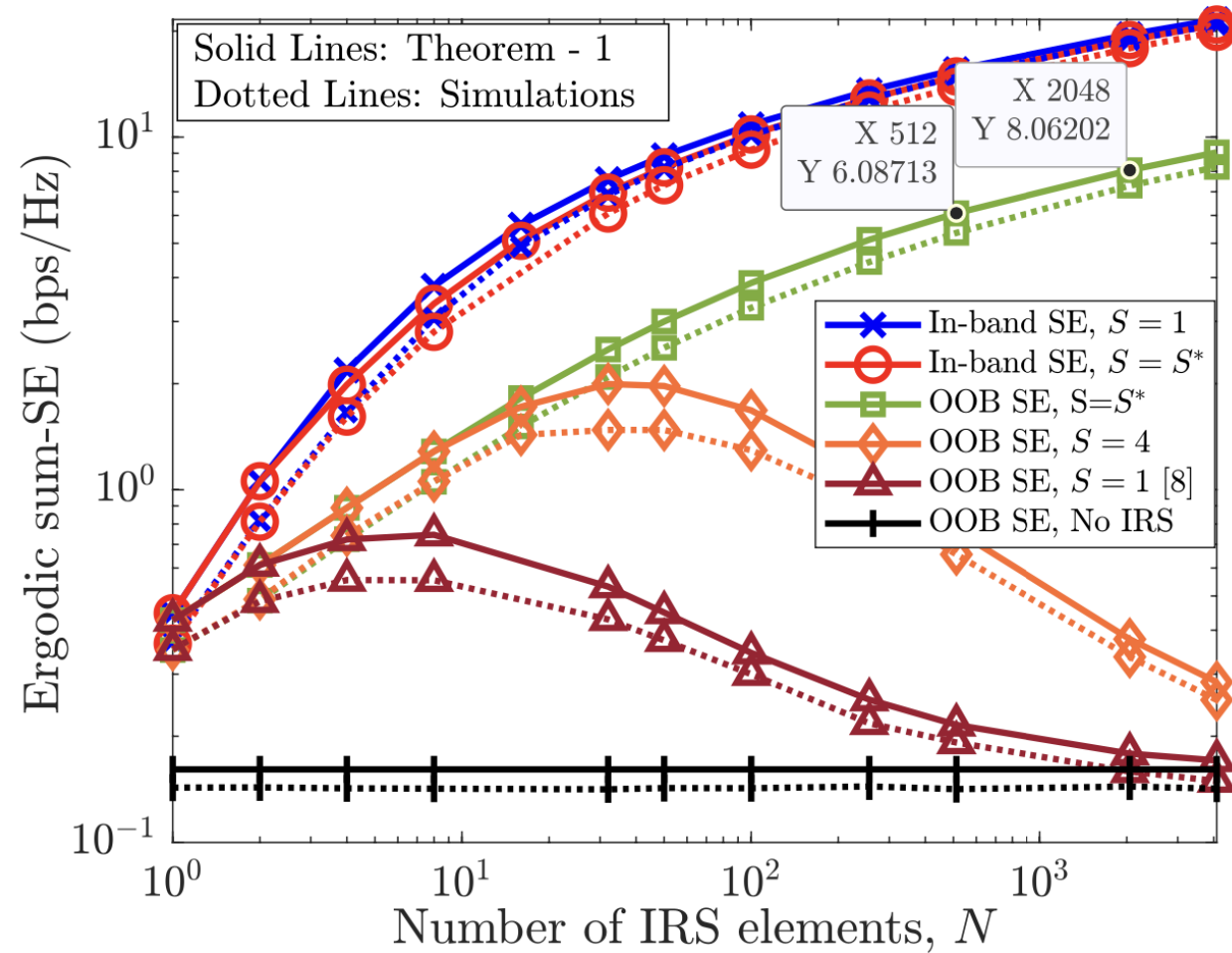
Main Result - 5: How to get $O(\log(N))$ SE?

Proposition 1. Consider a system where an MO deploys and controls multiple IRSs in a distributed fashion, using a total of N IRS elements. If the number of IRSs S each with M elements providing L cascaded paths at an OOB UE, satisfy

$$M \leq M^* \triangleq N^{\delta^*}, \text{ and, } S \geq S^* \triangleq \lceil N^{1-\delta^*} \rceil, \quad (19)$$

where $\delta^* = \min\{1, \log_N L\}$, then almost surely, the OOB SE attains the maximum scaling of $\mathcal{O}(\log_2(N))$ for any N, L .

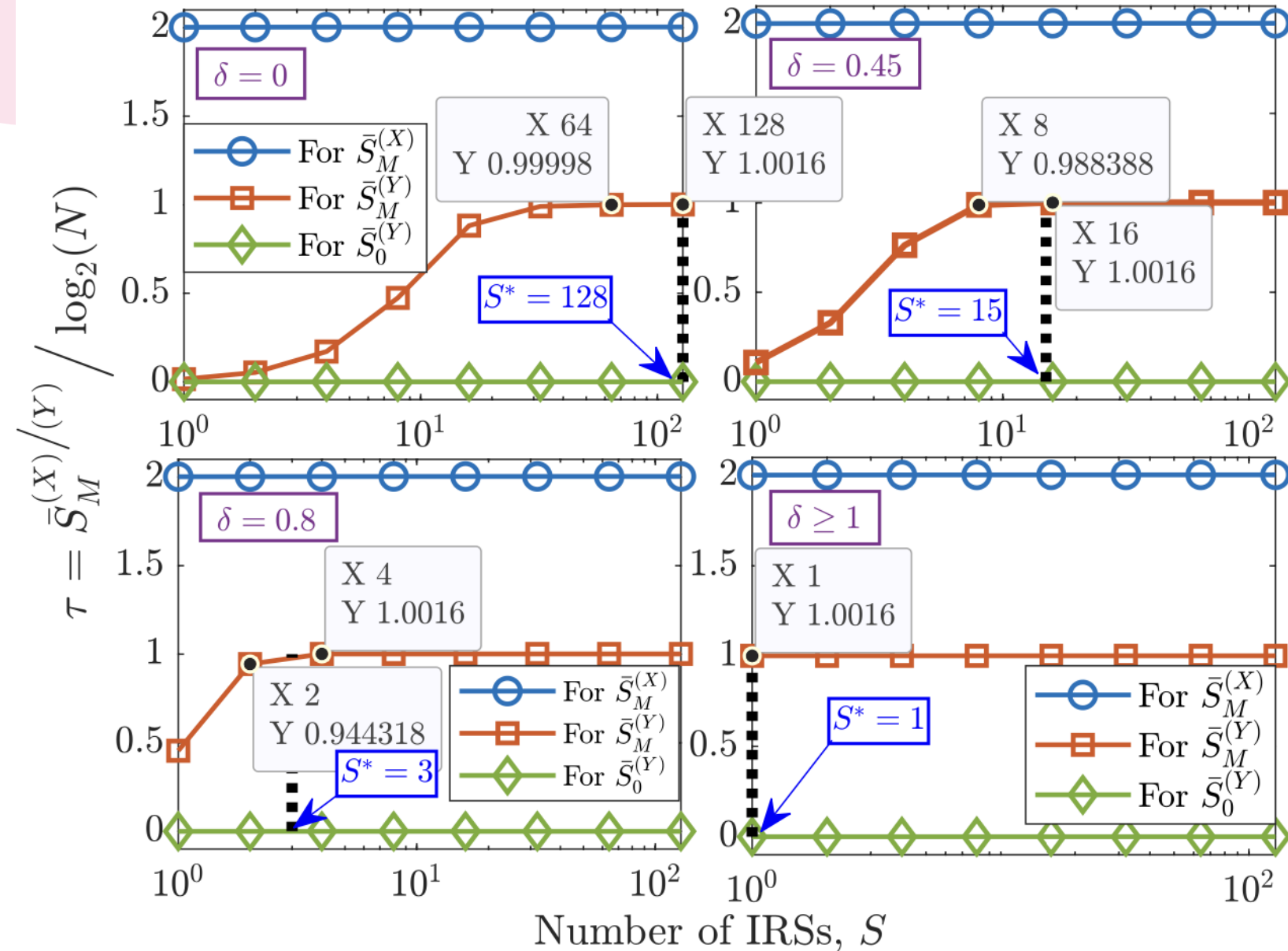
- With sufficiently many IRSs, OOB UEs
 - Almost surely obtains full scattering gain
 - $O(\log(N))$ SE gain like sub-6 GHz bands
- Distributed IRSs offer rich-scattering properties even in mmWave frequency bands! ☺



Assumptions:

- Negligible multi-IRS delay spread
- Same path-loss across sub-IRSs

How much rich-scattering is achieved?



➤ Notation:

$$\delta = \log_N L$$

➤ As the number of IRSs increases, the OOB SE is

$$\mathcal{O}(\tau \log_2(N)); \quad \tau \in [0, 1]$$

➤ With sufficiently many IRSs, the **maximum SE scaling** as that of **sub-6 GHz** bands is attained

Main Result - 6: Instant perks @ OOB UEs

➤ Outage Probability at OOB UE- q :

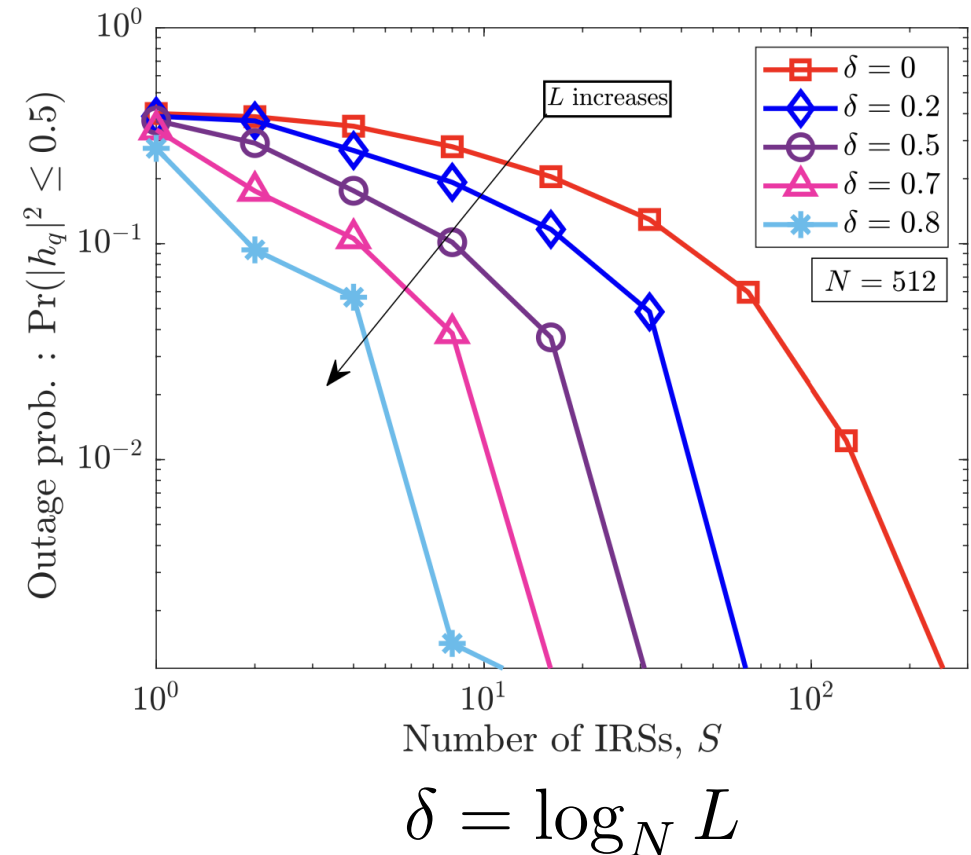
$$P_{out,q}^{\rho} = \left(1 - e^{-\rho/\beta_{d,q}}\right) \left(1 - \frac{\bar{L}}{M} \left(\frac{\bar{L} e^{\frac{\bar{L}\beta_{d,q}}{M^2\beta_{r,q}}}}{M^2\beta_{r,q}} \mathcal{I}_0\left(\rho; \beta_{d,q}, \frac{M^2}{\bar{L}}\beta_{r,q}\right) - e^{-\frac{\rho}{\beta_{d,q}}} \right) - e^{-\frac{\rho}{\beta_{d,q}}} \right)^S; \mathcal{I}_0(x; c_1, c_2) \triangleq \int_{c_1}^{\infty} e^{-\left(\frac{x}{t} + \frac{t}{c_2}\right)} dt.$$

➤ Outage prob. **reduces exponentially** in # IRSs

➤ Spatial diversity benefits @ OOB UEs for free!

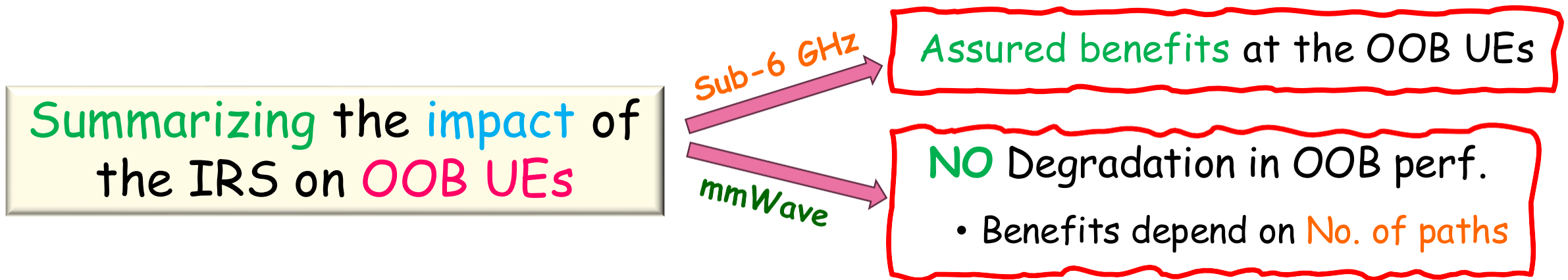
Distributed IRSs always **benefit**
every mobile operator, both in

1. **Average**, and
2. **Instantaneous senses** 😊



Concluding Remarks

- IRS **enhances** the average **SNR** by N^2 at (in-band) UE of operator X
- Operator Y gets **enhancement** of average **SNR** by **N** for **free of cost**
 - Creates additional channel links to the OOB UEs: Reception of **multiple** copies
- **Scalable results**: **NO** operator will be **disadvantaged** due to the IRS being controlled by one operator. **Every operator benefits!**
- **Distributed IRSs** **benefit** OOB operators **better** w/o affecting in-band performance



References

1. *L. Yashvanth*, and Chandra R. Murthy, **On the Impact of an IRS on the Out-of-Band Performance in Sub-6 GHz & mmWave Frequencies**, Accepted, IEEE Transactions on Communications (IEEE TCOM), May 2024.
2. *L. Yashvanth*, and Chandra R. Murthy, **Distributed IRSs Always Benefit Every Mobile Operator**, Accepted, IEEE Wireless Communications Letters (IEEE WCL), July 2024.
3. *L. Yashvanth*, and Chandra R. Murthy, **Cascaded Channel Estimation for Distributed IRS Aided mmWave Massive MIMO Systems**, Proc. IEEE Global Communications Conference (IEEE GLOBECOM '22), Rio de Janeiro, Brazil, Dec. 2022, pp. 717-723.

Thank You!

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